

WEIYI HUANG

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EDUCATION

Cornell University, College of Engineering, Bachelor of Science

Expected May 2027

Electrical and Computer Engineering, Aerospace Engineering Minor

Relevant Courses: Computer Architecture, Circuits, Microelectronics, Probability for Random Signals, Embedded Systems (SP25), Thermodynamics, Statics and Mechanics of Solids, Fluid Mechanics (SP25), Spaceflight Mechanics (SP25)

ENGINEERING EXPERIENCE

MAVRIK, Inc (startup), *Lead Electrical Engineer*

May 2025-Present

- Led embedded EE hardware leading up to and following **company acquisition** by HAVOC AI in February 2026.
- Owned electrical architecture of HATCHET UAS to win \$15k + SIBR at US Army's xTech Overwatch in October 2025.
- Architected Raspberry Pi compute module carrier board with focus of USB 3.0 signal integrity, external CV computing module PDM, and ethernet EMI mitigation while conforming to NDAA and specific SWaP-C constraints.
- Implemented and tested custom PCB breakout board to support modular payload release mechanisms, interfacing with STM32 and Cortex-M7-based microcontrollers through NVIDIA Jetson Nano companion computer.
- Developed low-level C++ code to interface payload board with custom companion software running Linux.
- Architected a redundant WAN failover solution using Starlink, DoodleLabs, and LTE through an AWS Lightsail instance to ensure persistent communication between GS and onboard companion computer.
- Implemented all flight-critical communication from Cube Orange, such as ESCs (CAN/PWM), uBlox GPS (I2C + UART), etc. Customized base set of MissionPlanner parameters to support rapid deployment of 10+ drones.

Cornell University AutoDrone, *Electrical Subteam Lead*

Sept 2024-Present

- Directed the implementation of all electrical systems for the high-speed "LawnDart" UAS, including RX/TX, FPV, and a high-discharge power architecture; performed extensive troubleshooting of electrical systems and INAV firmware to achieve reliable flight tests and consistent benchmarks for 250+ mph world record attempts.
- Engineered a custom motor test stand using an Arduino-based embedded system and supporting sensors to conduct rigorous performance evaluations, measuring stress force, RPM, and EMI noise under maximum throttle conditions.
- Appointed as Electrical Subteam Lead to lead architecture of 6+ UAS drones, based on prior success in reliable design.

Cornell University Biomedical Device (CUBMD), *Product Development and Media Subteam Member*

Feb-Dec 2024

- Developed a wireless pump control prototype using an Arduino-based embedded solution, enabling app integration via Bluetooth connectivity with the HC-08 Bluetooth module; iterated on prototype design by engineering a custom PCB to integrate all components, supporting an ATiny85 microcontroller in a condensed form factor.
- Generated 5,000+ new impressions and increased team applications by +25% through streamlining website redesign.

ENGINEERING EXPERIENCE

Multi-Core RISC-V Processor, *(Personal Project; Computer Architecture Final Project)*

August 2024-December 2024

- Developed hardware and software for multi-core RISC-V architecture, creating custom ring network interconnect with dedicated routing and arbitration units to support thread-level parallelism through multi-threaded C programs.

FET Amplifier Design and Optimization, *(Personal Project; Microelectronics Final Project)*

January 2025-June 2025

- Designed and constructed a cascaded FET amplifier and cascode amplifier using ALD1105 chips; characterized amplifier performance by measuring voltage gain, output resistance, and maximum voltage swing to ensure transistors remained in the saturation region and minimized distortion of input signal.

LEADERSHIP EXPERIENCE

Project Level the Field (Project LTF), *Chapter Founder and President*

May 2024-May 2025

- Founded Project LTF's Cornell chapter, a 501(c)(3) nonprofit providing college app resources to underserved students.
- Collaborated with a 7-member executive board, 11 mentors, and national chapter leadership to develop a 12-week strategic blueprint and execute 2 workshops, effectively sharing college app resources to 60+ underprivileged students.

SPECIALIZED SKILLS

Embedded Systems and Avionics: C/C++, Python, Verilog/SystemVerilog, Bash scripting; STM32, Cortex-M7, ATtiny, Arduino, Raspberry Pi; avionics buses (CAN, UART, SPI, I2C, PWM), sensor/actuator integration, EMI mitigation, redundant and fault-tolerant power architectures, flight control electronics, telemetry and data links, motor controllers.

Electronics & Tools: KiCad, LTSpice/SPICE simulation, Quartus Prime; oscilloscopes, spectrum/logic analyzers, DMM; soldering, harness fabrication, rapid prototyping, EMI testing; CAD/FEA (SolidWorks, Onshape, ANSYS); Linux, Git/GitHub.